Manav Aghera

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SUMMARY

AI/ML engineering student (B.Tech CSE, expected May 2027) seeking Machine Learning Engineer and AI Engineer internship or trainee roles. Builds end-to-end machine learning and large language model (LLM) systems, including ensemble learning, multi-agent LLM pipelines, retrieval-augmented generation (RAG), and deployed full-stack applications. **Top 1% on two Kaggle competitions**; **Anthropic Academy-certified** in MCP and RAG.

EXPERIENCE

Brainy Beam Technologies **Ahmedabad, India**
Data Science & Machine Learning Intern *Jul 2023 - Aug 2023*

- Built an end-to-end ML pipeline on a 40,000+ record banking dataset; the best of Logistic Regression, Decision Tree, and Random Forest models reached **92% accuracy** (evaluated with F1 and AUC-ROC).
- Cut data preprocessing time by **~30%** with automated cleaning scripts; findings informed campaign targeting.

Riot Games **Remote**
Beta Tester *Mar 2024 - Mar 2025*

- Play-tested **50+ pre-release builds** and wrote reproducible bug reports used for pre-launch fixes.

PROJECTS

Student Health Risk Engine | *Python, LightGBM, XGBoost, CatBoost, Optuna* *github.com/manavaghera/Predicting-Student-Health-Risk*

- Built a 3-model gradient-boosting ensemble (ensemble learning) for multi-class classification on 690K rows, with a heavily imbalanced target (86% majority class).
- Reached **0.952 balanced accuracy** in 5-fold cross-validation (0.950 on the public leaderboard) using per-fold target encoding, stacking, threshold optimization, and pseudo-labeling.
- Raised a 0.876 baseline to 0.952, keeping each technique only when cross-validation improved.

NewsSense: AI News Platform | *FastAPI, React 19, TypeScript, PostgreSQL, Docker* *github.com/manavaghera/LLM-NEWS-*

- Built a full-stack news platform with a FastAPI REST API and an hourly pipeline that ingests RSS and official sources, deduplicates them, checks LLM-generated claims against evidence, and publishes atomically.
- Added failure safeguards: editorial hold for failed claims, signed storage manifests with hash checks, and a database lease that prevents duplicate runs.
- Deployed on free-tier infrastructure (Cloudflare Pages, Render, Supabase) with GitHub Actions automation, **211 backend tests** (pytest), and Playwright end-to-end tests.

AutonomousContent: Multi-Agent Blog System | *LangGraph, Python, OpenRouter* *github.com/manavaghera/Multi-agent-blog*

- Built a LangGraph pipeline of research, writer, and validator agents; the validator runs a 4-stage quality check and sends weak drafts back for revision.
- Published posts scored **82-89/100** on automated validation; added self-healing with exponential backoff, model fallback, and a live pipeline dashboard. Ran entirely on free-tier LLM APIs at zero cost.

Clip Studio: Twitch Analytics & Short-Form Video Tool | *Python, Streamlit, Whisper, Ollama, ffmpeg* *github.com/manavaghera/Twitch-Bulk-Downloader*

- Built a Streamlit app that ranks Twitch clips by views-per-hour and generates vertical Shorts with burned-in captions.
- Used faster-whisper for transcription and an optional local LLM (Ollama) to rate highlight moments; GPU-accelerated encoding (NVENC) with CPU fallback. Includes a daily automation pipeline and **127 automated tests**.

CHRONUS: Memory-Grounded AI Research Platform | *Sentence-BERT, ChromaDB, Homomorphic Encryption* *Research in progress*

- Designing a privacy-preserving evolution of a local RAG pipeline (Sentence-BERT and ChromaDB) toward encrypted vector similarity search using additively homomorphic encryption.
- Designed offline research sandboxes for benchmarking Quantum NLP and spiking neural network approaches as possible future retrieval architectures.

TECHNICAL SKILLS

Machine Learning / Deep Learning: Python, PyTorch, TensorFlow, Scikit-learn, LightGBM, XGBoost, CatBoost, Optuna, feature engineering, model evaluation, cross-validation, computer vision (YOLOv8)
NLP / LLMs / Generative AI: Natural Language Processing (NLP), Large Language Models (LLMs), retrieval-augmented generation (RAG), LangGraph, LangChain, Prompt Engineering, Agentic Systems, Hugging Face, ChromaDB, Ollama
Software Engineering: FastAPI, REST APIs, React, TypeScript, Docker, GitHub Actions, PostgreSQL, SQL, Git, pytest, model deployment
Data: Pandas, NumPy, MLflow, data preprocessing

CERTIFICATIONS

- **Top 1%**, Kaggle “Playground Series S6E7” (3,355 teams); **Top 1%**, Kaggle “Autonomous Agent Prediction (Beta)” (570 teams)
- DeepLearning.AI, Machine Learning Specialization (with Stanford Online)
- DeepLearning.AI, TensorFlow Developer Professional Certificate
- Model Context Protocol (MCP) & RAG Certification, Anthropic Academy
- Microsoft Learn: Retrieval-Augmented Generation (RAG) Fundamentals; Introduction to Generative AI and Agents; Introduction to NLP Concepts
- Software Engineering Virtual Experience, JPMorgan Chase & Co. (Forage)
- Google Developer Student Clubs member; open-source contributions

Completed courses: DeepLearning.AI, Reinforcement Fine-Tuning of LLMs with GRPO; Stanford CS230: Deep Learning

EDUCATION

Bachelor of Technology (B.Tech), Computer Science Engineering, Parul University, Vadodara *Expected May 2027*
Diploma, Computer Engineering, NG Patel Polytechnic, Bardoli (CGPA 8.4) *Apr 2025*